

CALC-X

ADVANCED TERMINAL CALCULATOR

PROJECT REQUIREMENTS DOCUMENT (PRD)

A menu-driven mathematical problem-solving system developed in Python.

1. Problem Statement & Description

CALC-X is an advanced terminal-based mathematical calculator developed in Python to provide a single platform for solving different mathematical problems. The system will support BODMAS-based expression evaluation, scientific functions such as logarithms and trigonometry, algebra and equation solving, matrix operations, statistics, basic calculus, and calculation history. The goal is to provide these mathematical operations through one simple, menu-driven application.

2. Interface

2.1 How to Run

Run:

```
python main.py
```

Python 3 should be installed. The application runs in a terminal or command prompt.

2.2 Main Menu

Choice	Module
1	Basic Calculator
2	Scientific Calculator
3	Algebra Solver
4	Equation Solver
5	Matrix Operations
6	Statistics
7	Calculus
8	History
9	Exit

2.3 Inputs

The program accepts menu choices, mathematical expressions, numerical values, equations, matrix elements, statistics data, polynomial expressions, and history file choices.

2.4 Outputs

The program displays calculated answers, solved variables, matrices, statistical measures, calculus results, error messages, and calculation history.

3. Functional Requirements

ID	Requirement	Description
FR1	Main Menu and Navigation	Display menu choices 1-9, open modules 1-8, return after operations, and exit on 9.
FR2	Basic Calculator	Support +, -, *, /, %, parentheses, and BODMAS/operator precedence.
FR3	Scientific Calculator	Support powers, pow(), sqrt(), cbrt(), log(), ln(), logb(), trig/inverse trig, pi, e, abs(), floor(), ceil(), and degree/radian mode. Hyperbolic functions are excluded.
FR4	Full Expression Evaluation / BODMAS	Evaluate complete expressions with numbers, operators, parentheses, powers, constants, and scientific functions using mathematical precedence.
FR5	Algebra Solver	Solve linear equations, quadratic equations, and basic polynomial calculations using Python lists and functions.
FR6	Equation Solver	Solve simultaneous 2-variable/2-equation and 3-variable/3-equation systems and detect no-unique-solution cases.
FR7	Matrix Operations	Support addition, subtraction, multiplication, transpose, determinant, and inverse with dimension validation.
FR8	Statistics	Calculate mean, median, mode, minimum, maximum, range, variance, and standard deviation.
FR9	Calculus	Perform basic polynomial differentiation and integration.
FR10	Calculation History	Store successful calculations and support viewing, saving, and loading history.
FR11	Input Validation and Error Handling	Validate menus, expressions, parentheses, mathematical domains, matrices, and equations. Invalid input must not crash the program.

FR1 — Detailed Requirements

- Display main menu.
- Accept only 1-9.
- Open selected module.
- Return to menu.
- Exit on 9.

FR2 — Detailed Requirements

- Support +, -, *, /, %.
- Support parentheses.
- Apply BODMAS.

FR3 — Detailed Requirements

- Support ^ and pow().
- Support sqrt() and cbrt().
- Support log(), ln(), logb().
- Support sin(), cos(), tan(), asin(), acos(), atan().
- Support pi and e.
- Support abs(), floor(), ceil().
- Support degree/radian modes.
- Exclude hyperbolic functions.

FR4 — Detailed Requirements

- Accept complete expressions.
- Identify numbers, operators, parentheses, constants, and functions.
- Evaluate grouped expressions/functions.
- Apply powers/roots, then $\times/\div/\%$, then $+/-$.
- Return errors for invalid expressions or undefined operations.

FR5 — Detailed Requirements

- Solve $2x + 5 = 15$.
- Solve $x^2 - 5x + 6 = 0$.
- Support polynomial calculations.
- Report repeated and non-real roots appropriately.

FR6 — Detailed Requirements

- Solve 2×2 systems.
- Solve 3×3 systems.
- Detect systems without a unique solution.

FR7 — Detailed Requirements

- Add/subtract equal-size matrices.
- Multiply compatible matrices.
- Transpose.

- Determinant of square matrices.
- Inverse of non-singular square matrices.

FR8 — Detailed Requirements

- Accept numerical data.
- Calculate mean, median, mode, min, max, range, variance, standard deviation.
- Handle repeated and negative values.

FR9 — Detailed Requirements

- Differentiate basic polynomials.
- Integrate basic polynomials.
- Display + C.

FR10 — Detailed Requirements

- Store successful calculations.
- View history.
- Save history.
- Load history.

FR11 — Detailed Requirements

- Reject invalid menus.
- Reject malformed expressions/parentheses.
- Prevent division/modulus by zero.
- Validate math domains.
- Validate matrix dimensions.
- Prevent crashes after repeated invalid input.

3.1 Scientific Expression Examples

Expression	Expected Result	Notes
$\sqrt{144} + \log(1000) * \sin(30)$	13.5	Degree mode
$2 + 5 * (3 + 4)$	37	BODMAS
$\sqrt{144} + \log(100) * 2$	16	Base-10 log
$\sin(30) + \cos(60)$	1	Degree mode
$\log_b(81,3) + 5$	9	Base-3 logarithm
$((5+3)^2 + \sqrt{81})/\log(10)$	73	Base-10 log

Expression	Expected Result	Notes
$((5+3) * 2^3 + \log(1000))/\text{sqrt}(16)$	16.75	Base-10 log

4. Edge Cases

ID	Condition	Expected Handling
EC01	Menu choice 0	Invalid choice; no crash.
EC02	Menu choice 10	Invalid choice; no crash.
EC03	Menu input abc	Reject non-numeric input.
EC04	Blank menu input	Reject blank input.
EC05	Menu input 1.5	Reject non-integer choice.
EC06	Empty expression	Reject and request valid input.
EC07	Invalid letters in expression	Reject unsupported tokens.
EC08	Invalid operator	Reject malformed operator sequence.
EC09	Missing closing parenthesis	Report unmatched parenthesis.
EC10	Extra closing parenthesis	Report unmatched parenthesis.
EC11	Division by zero	Show mathematical error.
EC12	Modulus by zero	Show mathematical error.
EC13	$\text{sqrt}(-9)$	Reject invalid domain.
EC14	$\log(0)$	Reject undefined logarithm.
EC15	$\log(-10)$	Reject invalid domain.
EC16	$\ln(0)$	Reject undefined logarithm.
EC17	$\log_b(10,1)$	Reject base 1.
EC18	$\tan(90)$ degree mode	Detect undefined/near-undefined tangent.
EC19	Invalid trig mode	Reject unsupported mode.
EC20	Very large number	Handle range or report overflow.
EC21	Decimal where integer required	Reject or request integer.
EC22	Linear equation with zero x coefficient	Detect invalid/non-x case.

ID	Condition	Expected Handling
EC23	Quadratic discriminant 0	Return repeated real root.
EC24	Quadratic negative discriminant	Report non-real roots / unsupported real-only result.
EC25	Simultaneous equations no unique solution	Report no unique solution.
EC26	Matrix addition incompatible dimensions	Reject.
EC27	Matrix multiplication incompatible dimensions	Reject.
EC28	Determinant non-square	Reject.
EC29	Inverse non-square	Reject.
EC30	Inverse singular	Report no inverse.
EC31	Statistics zero values	Reject empty dataset.
EC32	Statistics one value	Return valid applicable statistics.
EC33	Negative statistics values	Process valid data.
EC34	Malformed calculus polynomial	Reject malformed polynomial.
EC35	Empty history	Show empty-history message.
EC36	Missing history file	Show file-not-found/load error.
EC37	History file cannot be opened	Show file access error.
EC38	Multiple invalid inputs consecutively	Continue safely and allow retry.
EC39	Exit before calculation	Exit normally.

5. Test Cases

Menu & Navigation

Test ID	FR	Input / Action	Expected Result
TC01	FR1	Start program	Main menu displayed.
TC02	FR1	Enter 1-8	Correct module opens.
TC03	FR1	Enter 9	Program exits.

Basic Calculator

Test ID	FR	Input / Action	Expected Result
TC04	FR2	25 + 10	35.
TC05	FR2	25 - 10	15.
TC06	FR2	25 * 10	250.
TC07	FR2	25 / 5	5.
TC08	FR2	25 % 4	1.
TC09	FR2	(25 + 10) * 2	70.
TC10	FR2	25 + 10 * 2	45.
TC11	FR2	10 / 0	Error; no crash.

Scientific Calculator

Test ID	FR	Input / Action	Expected Result
TC12	FR3	2^5	32.
TC13	FR3	sqrt(144)	12.
TC14	FR3	cbrt(27)	3.
TC15	FR3	log(1000)	3.
TC16	FR3	ln(e)	1.
TC17	FR3	logb(81,3)	4.
TC18	FR3	sin(30), degree	0.5.
TC19	FR3	cos(60), degree	0.5.
TC20	FR3	tan(45), degree	1.
TC21	FR3	asin(0.5), degree	30°.
TC22	FR3	acos(0.5), degree	60°.
TC23	FR3	atan(1), degree	45°.
TC24	FR3	pi and e	Correct constants.
TC25	FR3	abs(-8), floor(3.9), ceil(3.1)	8, 3, 4.
TC26	FR3	sin(1.570796), radian	Approximately 1.
TC27	FR3	Hyperbolic function	Reject as unsupported.

BODMAS / Full Expression Evaluation

Test ID	FR	Input / Action	Expected Result
TC28	FR4	$2 + 5 * (3 + 4)$	37.
TC29	FR4	$\text{sqrt}(144) + \log(1000) * \sin(30)$, degree	13.5.
TC30	FR4	$\text{sqrt}(144) + \log(100) * 2$	16.
TC31	FR4	$\sin(30) + \cos(60)$, degree	1.
TC32	FR4	$\log_b(81, 3) + 5$	9.
TC33	FR4	$((5+3)^2 + \text{sqrt}(81)) / \log(10)$	73.
TC34	FR4	$((5+3) * 2^3 + \log(1000)) / \text{sqrt}(16)$	16.75.
TC35	FR4	$2 + 3 * 4$	14, not 20.
TC36	FR4	$(2 + 3) * 4$	20.
TC37	FR4	Malformed expression	Error; no crash.

Algebra Solver

Test ID	FR	Input / Action	Expected Result
TC38	FR5	$2x + 5 = 15$	$x = 5$.
TC39	FR5	$x^2 - 5x + 6 = 0$	$x_1 = 2, x_2 = 3$.
TC40	FR5	$x^2 - 4x + 4 = 0$	Repeated root $x = 2$.
TC41	FR5	$x^2 + 1 = 0$	Report non-real roots / unsupported real-only result.
TC42	FR5	Malformed algebra	Error; no crash.

Equation Solver

Test ID	FR	Input / Action	Expected Result
TC43	FR6	$2x + 3y = 13; 4x - y = 5$	$x=2, y=3$.
TC44	FR6	Valid 3×3 system	Correct unique solution.
TC45	FR6	Dependent equations	No unique solution.
TC46	FR6	Inconsistent equations	No solution.

Test ID	FR	Input / Action	Expected Result
TC47	FR6	Malformed system	Error; no crash.

Matrix Operations

Test ID	FR	Input / Action	Expected Result
TC48	FR7	[[1,2],[3,4]] + [[5,6],[7,8]]	[[6,8],[10,12]].
TC49	FR7	Matrix subtraction	Correct result.
TC50	FR7	Matrix multiplication	Correct result.
TC51	FR7	Transpose [[1,2],[3,4]]	[[1,3],[2,4]].
TC52	FR7	Det([[1,2],[3,4]])	-2.
TC53	FR7	Inverse [[1,2],[3,4]]	Correct inverse.
TC54	FR7	Incompatible addition/multiplication	Reject.
TC55	FR7	Non-square determinant/inverse	Reject.
TC56	FR7	Singular inverse	Report no inverse.

Statistics

Test ID	FR	Input / Action	Expected Result
TC57	FR8	10 20 20 30 40	Mean 24; median 20; mode 20; min 10; max 40; range 30.
TC58	FR8	1 2 3 4 5	Mean 3; median 3.
TC59	FR8	Repeated values	Correct mode.
TC60	FR8	Negative values	Correct statistics.
TC61	FR8	One value: 10	Valid applicable statistics.
TC62	FR8	Empty data	Reject input.

Calculus

Test ID	FR	Input / Action	Expected Result
TC63	FR9	$3x^2 + 4x + 5$ derivative	$6x + 4$.
TC64	FR9	$6x + 4$ integral	$3x^2 + 4x + C$.

Test ID	FR	Input / Action	Expected Result
TC65	FR9	5 derivative	0.
TC66	FR9	Polynomial with several terms	Correct derivative/integral.
TC67	FR9	Malformed polynomial	Reject.

History

Test ID	FR	Input / Action	Expected Result
TC68	FR10	Successful calculation	Stored.
TC69	FR10	View history	Entries displayed.
TC70	FR10	Save history	File created/updated.
TC71	FR10	Load history	Entries restored.
TC72	FR10	Empty history	Empty-history message.
TC73	FR10	Missing/inaccessible history file	Appropriate file error.

General Validation

Test ID	FR	Input / Action	Expected Result
TC74	FR11	Menu 0 or 10	Invalid choice; no crash.
TC75	FR11	Menu abc / blank	Invalid input; no crash.
TC76	FR11	Unmatched parentheses	Error; no crash.
TC77	FR11	$\sqrt{-9}$	Domain error; no crash.
TC78	FR11	$\log(0)$ / $\ln(0)$	Domain error; no crash.
TC79	FR11	Repeated invalid inputs	Program remains usable.
TC80	FR11	Exit before calculation	Program exits normally.

6. Functional Requirement → Test Coverage

Requirement	Test Cases	Coverage
FR1	TC01-TC03	Menu and navigation
FR2	TC04-TC11	Basic calculator

Requirement	Test Cases	Coverage
FR3	TC12-TC27	Scientific calculator
FR4	TC28-TC37	BODMAS/expression evaluation
FR5	TC38-TC42	Algebra
FR6	TC43-TC47	Equations
FR7	TC48-TC56	Matrices
FR8	TC57-TC62	Statistics
FR9	TC63-TC67	Calculus
FR10	TC68-TC73	History
FR11	TC74-TC80	Validation/error handling

7. Core Implementation Note

The main advanced feature of CALC-X is full expression evaluation. The intended processing flow is:

INPUT → VALIDATE → IDENTIFY NUMBERS / OPERATORS / FUNCTIONS → HANDLE BRACKETS / GROUPED EXPRESSIONS → EVALUATE FUNCTIONS → POWERS / ROOTS → $\times$ / $\div$ / $\%$ → $+$ / $-$ → ANSWER

The implementation can use Python strings, lists, dictionaries, variables, and functions. Lists can hold tokens, numbers, operators, and intermediate results. Python functions can separate calculator modules without requiring object-oriented programming.

8. Scope Summary

Scope	Items
Included	Basic calculator; scientific calculator; BODMAS expression evaluation; algebra; equations; matrices; statistics; basic calculus; history.
Excluded	Number System Converter; Financial Calculator; Permutation; Combination; Basic Probability; Binomial Probability; Complex Numbers; Hyperbolic Functions.